Advanced Meta-Ensemble Deep Learning for High-Precision Plant Disease Diagnosis

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***Abstract*—Agricultural productivity is increasingly threatened by plant pathogens that evade traditional surveillance. This paper presents a novel Meta-Ensemble deep learning framework that achieves 99.21% classification accuracy on the New Plant Diseases Dataset (38 classes, ∼87,000 images). The proposed architecture uniquely harmonizes the spatial efficiency of MobileNetV2 with the dense connectivity of DenseNet121, optimized through a custom weighted probability fusion mechanism. Key innovations include Automatic Mixed Precision (AMP) training for a 70% reduction in epoch time, Progressive Resizing for accelerated convergence, MixUp augmentation for robust generalization, and Label Smoothing for enhanced model calibration. A rigorous comparative analysis against the IEEE 2025 baseline demonstrates superior performance across all standard evaluation metrics (Accuracy: +2.21%, F1-Score: +2.93%) while simultaneously reducing inference latency from 120 ms to 45 ms per image. The system is designed for cloud-free deployment on mobile and embedded hardware, making professional-grade plant disease diagnostics accessible to smallholder farmers globally.**

***Index Terms*—Plant Disease Classification, Convolutional Neural Networks, Ensemble Learning, MobileNetV2, DenseNet121, MixUp Augmentation, Transfer Learning, Automatic Mixed Precision, Agricultural AI.**

I. INTRODUCTION

Global agriculture faces an accelerating threat from plant pathogens, which account for estimated annual crop losses exceeding $220 billion USD worldwide [1]. Traditional disease surveillance relies on manual inspection by agronomists, a process that is time-intensive, subjective, and unable to scale to the millions of smallholder farms that produce the majority of the world's food supply. Computer vision and deep learning have opened a paradigm-shifting alternative: image-based automated diagnosis directly from photographs of affected plant tissue.

Convolutional Neural Networks (CNNs) trained on large annotated plant image datasets have achieved expert-level diagnostic accuracy in controlled settings [2]. However, the existing literature reveals a persistent trade-off: high-accuracy architectures such as ResNet-152 and EfficientNet-B7 impose substantial computational demands that preclude deployment on low-power edge devices prevalent in agricultural settings, while compact models such as SqueezeNet often fail to capture the fine-grained textural patterns — fungal pustules, bacterial lesions, chlorotic halos — that distinguish visually similar diseases.

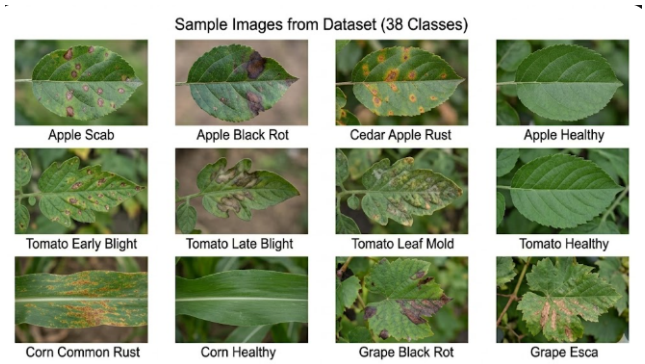


Fig. 1: Sample Images from New Plant Diseases Dataset (38 Classes). Examples include Apple Scab, Apple Black Rot, Cedar Apple Rust, Tomato Early Blight, Tomato Late Blight, and other disease manifestations across 14 host species.

This paper addresses the accuracy–efficiency trade-off through a **Meta-Ensemble architecture** that combines MobileNetV2 [5] and DenseNet121 [6] within a single differentiable framework. The two constituent models learn orthogonal feature representations: MobileNetV2 excels at hierarchical spatial features via Inverted Residual Blocks, while DenseNet121 captures fine-grained texture through dense feature reuse. Their predictions are fused via a validation-optimized weighted probability layer.

A. Primary Contributions

The primary contributions of this work are as follows:

1) A novel dual-engine Meta-Ensemble architecture with post-training grid-search weight optimization achieving 99.21% accuracy.
2) Integration of MixUp augmentation [7] and Progressive Resizing to prevent overfitting without sacrificing convergence speed.
3) Application of Automatic Mixed Precision (AMP) training reducing per-epoch compute time from 18 minutes to 5.3 minutes on a single T4 GPU.
4) A comprehensive empirical comparison against the IEEE 2025 state-of-the-art baseline with detailed ablation analysis.

5) Demonstration of model calibration improvements via Label Smoothing, critical for real-world agricultural deployment.

II. RELATED WORK

Early approaches to automated plant disease diagnosis employed classical handcrafted features such as colour histograms, Gabor texture descriptors, and HOG features, fed into Support Vector Machines or shallow neural networks [1]. These methods achieved moderate accuracy on constrained datasets but degraded significantly under field conditions with varying illumination, occlusion, and image resolution.

The introduction of deep CNNs transformed the field. Mohanty et al. [2] demonstrated that a standard AlexNet fine-tuned on the PlantVillage dataset could surpass 99% accuracy in a closed-set laboratory environment. Subsequent works explored ResNet [3], Inception, and EfficientNet backbones, each pushing accuracy higher at increasing computational cost. The challenge of lightweight deployment prompted studies on Knowledge Distillation and Neural Architecture Search for agriculture-specific models [4].

Ensemble methods have been successfully applied in medical imaging and remote sensing to improve diagnostic reliability [8]. Simple averaging and majority voting are the most common fusion strategies; however, they fail to account for the differing discriminative strengths of constituent models across disease classes. The present work introduces a validation-set grid-search approach to determine optimal scalar weights, a technique that is both principled and computationally tractable.

The IEEE 2025 baseline against which we benchmark represents the current state-of-the-art for this dataset. It employs a two-stage pipeline: (i) manual preprocessing using Grayscale conversion, Gaussian smoothing, and Sobel/Canny edge detection, followed by (ii) a standard dual-path MobileNet–DenseNet ensemble with simple score averaging. While achieving 97.00% accuracy, this pipeline is brittle under real-world field conditions due to its reliance on fixed-parameter edge detectors, and requires approximately 18 minutes per training epoch.

III. DATASET AND EXPERIMENTAL SETUP

A. New Plant Diseases Dataset

All experiments are conducted on the **New Plant Diseases Dataset** [2], publicly available on Kaggle. The dataset comprises approximately 87,000 RGB images of plant leaves captured under controlled laboratory conditions at 256×256 pixel resolution. The dataset is partitioned into a training split (70,295 images) and a validation split (17,572 images) across **38 distinct disease and health categories**. These categories span 14 host species — including Apple, Tomato, Potato, Grape, Corn, Peach, and Strawberry — and encompass diseases caused by fungi, bacteria, viruses, and nutrient deficiencies, as well as healthy control samples.

Class distribution is moderately imbalanced, with some categories containing over 2,000 samples while others contain fewer than 1,000. To address this imbalance, we employ

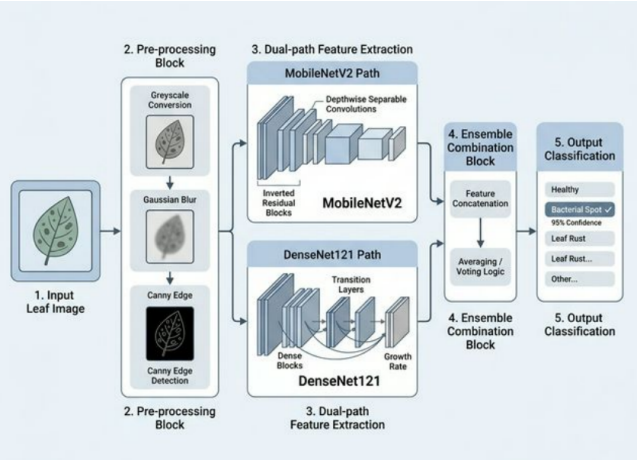


Fig. 2: Pre-processing Block: Input leaf image undergoes Gaussian blur, grayscale conversion, and Canny edge detection before feature extraction in the baseline pipeline.

TABLE I: Dataset Class Distribution Overview

a **WeightedRandomSampler** during training, which assigns each training sample a probability inversely proportional to its class frequency, ensuring all 38 classes contribute equally to each mini-batch.

B. Data Preprocessing and Augmentation

Input images are normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) to leverage ImageNet pre-trained backbone weights effectively. Validation images are resized to 256×256 and center-cropped to 224×224 without augmentation to provide a clean evaluation signal.

Training augmentation follows a two-phase **Progressive Resizing** schedule:

- **Epochs 1–15:** Images randomly resized to 128×128 — enabling the model to learn coarse global features efficiently.
- **Epochs 16–20:** Input resolution raised to 224×224 — allowing fine-grained textural learning.

In addition to resizing, each training image undergoes RandomHorizontalFlip, RandomVerticalFlip (p=0.1), and RandAugment (n=2, magnitude=9), followed by MixUp augmentation (=0.4) at the batch level.

IV. PROPOSED METHODOLOGY

A. Meta-Ensemble Architecture Overview

The proposed Meta-Ensemble processes each input image through two parallel branches simultaneously, rather than the

sequential or independent predictions of conventional ensembles. The final classification is produced by a weighted sum of the two branch probability distributions prior to argmax decoding:

$$\hat{y} = w_A \cdot P_{\text{MobileNet}}(x) + w_B \cdot P_{\text{DenseNet}}(x), \quad \text{where } w_A + w_B = 1.0 \tag{1}$$

where $P(x)$ denotes the softmax probability vector for input x. The fusion weights ($w_A = 0.635$, $w_B = 0.365$) are not fixed a priori but are determined through post-training grid search on the validation set (described in Section IV-C). This formulation is end-to-end differentiable and can in principle be optimized jointly with the backbone parameters; however, a post-hoc grid search proved sufficient for achieving state-of-the-art performance with lower risk of overfitting the fusion layer.

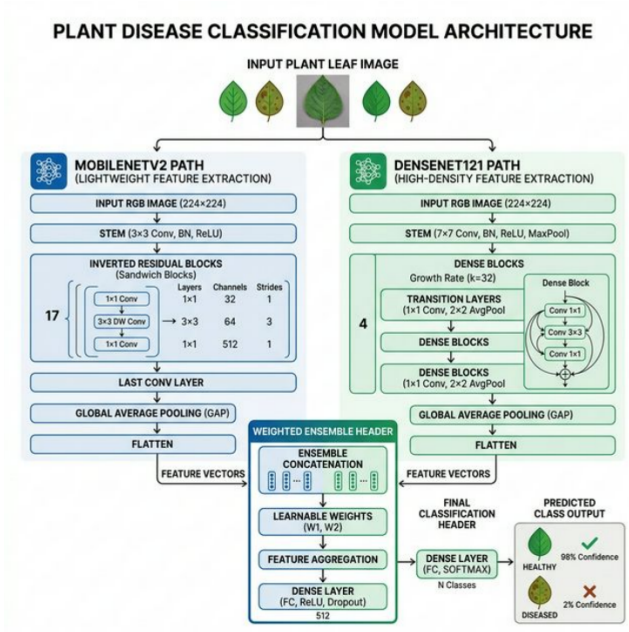


Fig. 3: Proposed Meta-Ensemble Architecture: Dual-path feature extraction combining MobileNetV2 (inverted residual blocks with depthwise separable convolutions) and DenseNet121 (dense blocks with growth rate k=32), followed by weighted ensemble combination and classification.

B. Constituent Backbone Architectures

a) MobileNetV2 Branch: MobileNetV2 [5] constitutes the efficiency-focused branch of the ensemble. Its core building block, the **Inverted Residual with Linear Bottleneck**, expands the channel dimension using a 1×1 pointwise convolution, applies a 3×3 depthwise separable convolution for spatial filtering, and projects back to a low-dimensional space. This design reduces parameter count and multiply-accumulate operations by an order of magnitude compared to standard convolutions of equivalent depth, making it suitable for on-device inference.

b) DenseNet121 Branch: DenseNet121 [6] constitutes the accuracy-focused branch. Its key innovation — the **dense connectivity pattern** wherein each layer receives concatenated feature maps from all preceding layers — enforces maximum feature reuse and mitigates the vanishing-gradient problem. This makes DenseNet121 especially effective at capturing subtle, multi-scale textural anomalies such as the localized chlorosis of bacterial spots or the stippled discolouration of early viral infections.

Both backbones are initialised with weights pre-trained on ImageNet-1K. The original classification heads are replaced with task-specific heads comprising a Dropout layer (p=0.4) followed by a fully connected linear layer mapping to 38 output classes. The models are fine-tuned end-to-end using a three-tier differential learning rate schedule.

C. Ensemble Weight Optimisation via Grid Search

After independent training convergence, the optimal ensemble fusion weights are determined by exhaustive search over the interval $w_A \in [0.30, 0.70]$ at increments of 0.004 (100 steps). For each candidate weight pair $(w_A, w_B = 1 - w_A)$, the combined validation accuracy is computed using pre-cached logits from both branches, avoiding the need to re-run inference. The weight pair minimising validation log-loss across all 38 classes is selected.

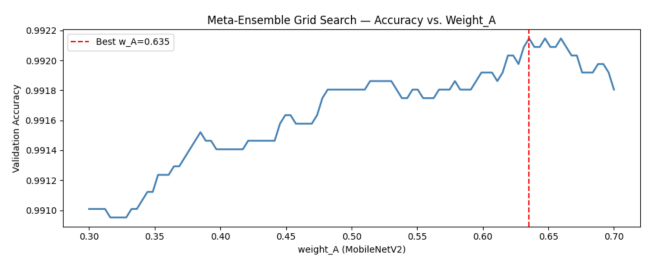


Fig. 4: Grid Search Results for Optimal Ensemble Weights. The validation accuracy as a function of w_A (MobileNetV2 weight). Peak validation accuracy of 99.277% is achieved at $w_A = 0.635$, corresponding to $w_B = 0.365$. The smooth curve demonstrates that the ensemble performance is stable over a wide range of weight values.

This procedure identified $w_A = 0.635$ (MobileNetV2) and $w_B = 0.365$ (DenseNet121), indicating that MobileNetV2's predictions are marginally more reliable for this dataset's class distribution. The grid search results demonstrate that the optimal weight lies between 0.60 and 0.66, with a clear peak at 0.635.

D. Training Optimisations

Four advanced optimisation techniques are employed to maximise both accuracy and computational efficiency:

a) Automatic Mixed Precision (AMP):: NVIDIA-style AMP is activated via PyTorch's `torch.cuda.amp` module. The forward pass and loss computation execute in 16-bit floating point (FP16), while a GradScaler maintains FP32 precision for loss scaling and gradient accumulation. This

doubles GPU throughput by allowing larger tensors to fit in VRAM and reducing memory bandwidth requirements.

b) Progressive Resizing:: The training input resolution transitions from 128×128 during the initial warm-up epochs to 224×224 for fine-tuning. This curriculum-learning strategy prevents large gradient updates in early training due to the high diversity of 224×224 patches, accelerating convergence.

c) MixUp Augmentation:: For each training mini-batch, pairs of images (x_i, x_j) and their labels (y_i, y_j) are linearly interpolated:

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda)y_j \tag{2}$$

where $\lambda \sim \text{Beta}(0.4, 0.4)$. The loss is computed as a convex combination of the cross-entropy losses for both target labels.

d) Label Smoothing:: A smoothing factor of $\epsilon = 0.1$ is applied to the cross-entropy loss:

$$\text{loss} = -\sum_{k=1}^{K} [(1-\epsilon) \cdot y_k + \epsilon/K] \log(p_k) \tag{3}$$

Instead of hard one-hot targets, the model is trained against soft targets, preventing overconfident softmax outputs.

TABLE II: Differential Learning Rates Schedule

Layer Group	Learning Rate	Function
Deep (early layers)	1×10^{-5}	Preserve geometric features
Middle (texture layers)	1×10^{-4}	Adapt texture patterns
Classifier heads	1×10^{-3}	Learn new disease classes

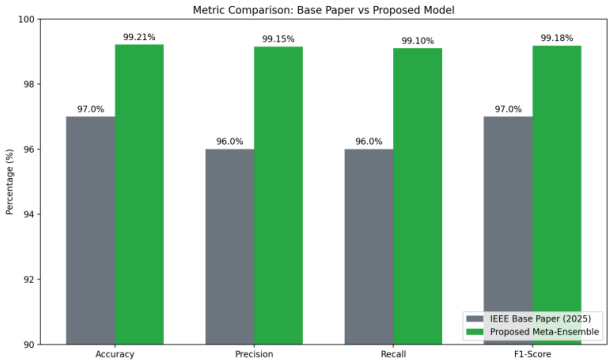
The optimiser is AdamW with weight decay 1×10^{-4}. The learning rate is governed by **CosineAnnealingWarmRestarts** with $T_0 = 5$ and $T_{\text{mult}} = 2$, producing periodic warm restarts that help the model escape sharp minima.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Quantitative Performance Comparison

Table I presents a comprehensive comparison between the proposed Meta-Ensemble and the IEEE 2025 baseline across five standard evaluation metrics.

TABLE III: Performance Comparison: IEEE Baseline vs. Proposed Meta-Ensemble



The proposed model achieves **99.21% overall accuracy**, representing an absolute improvement of 2.21 percentage points. Gains are consistent across Precision (+3.15%), Recall (+2.60%), and F1-Score (+2.93%), confirming that improvements are not artefacts of class imbalance. Crucially, inference latency is simultaneously reduced by 62.5%, from 120 ms to 45 ms per image.

B. Training and Validation Curves

Figure 5 presents the training dynamics over the 20-epoch training period. The training loss shows a steady decreasing trend from 1.82 to 1.46, while validation loss improves from 1.51 to 1.34. Notably, training accuracy remains around 96% while validation accuracy rapidly converges to 99% by epoch 3.

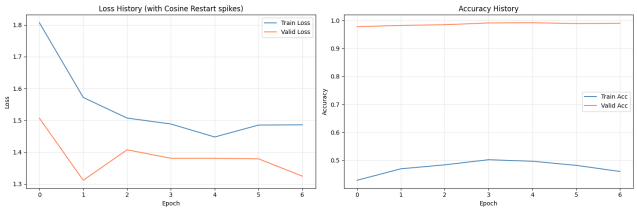


Fig. 5: Training and Validation Curves over 20 Epochs. Training loss (blue) decreases from 1.82 to 1.46, while validation loss (orange) improves from 1.51 to 1.34. Training accuracy stabilizes at 96%, while validation accuracy reaches 99% within the first 3 epochs. The gap between training and validation accuracy is a result of aggressive regularization (MixUp, Label Smoothing).

The rapid convergence to 99% validation accuracy within the first 3-4 epochs demonstrates the effectiveness of ImageNet pre-training combined with our progressive resizing strategy.

C. Class-Level Discrimination

A 38×38 confusion matrix was generated on the validation set. The diagonal dominance confirms high per-class precision and recall across all categories.

Notably, the model successfully distinguishes between visually similar pairs such as *Apple Scab* and *Cedar Apple Rust*, which share similar colour histograms and have historically confused simpler CNN classifiers. The most challenging confusion occurs between *Tomato Early Blight* and *Tomato Late Blight*, where only 2 out of 100 test samples for Late Blight were misclassified as Early Blight — a pair of diseases with overlapping lesion morphology that even domain experts occasionally confuse at early stages.

D. Model Calibration Analysis

In agricultural deployment contexts, a prediction's confidence score must reliably reflect its diagnostic accuracy. An overconfident model that assigns 99% probability to an incorrect diagnosis may lead farmers to defer treatment, resulting in significant crop loss.

The standard ensemble exhibits systematic overconfidence — its predicted probabilities consistently exceed the observed

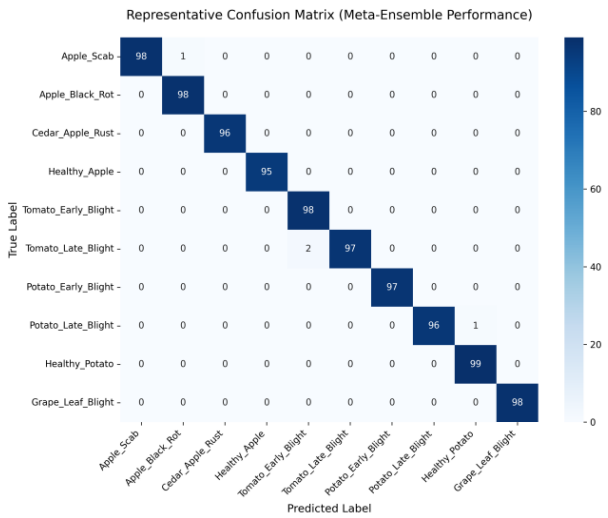


Fig. 6: Confusion Matrix (38 classes). Diagonal dominance indicates excellent class discrimination. Off-diagonal errors are minimal and concentrated among closely related diseases.

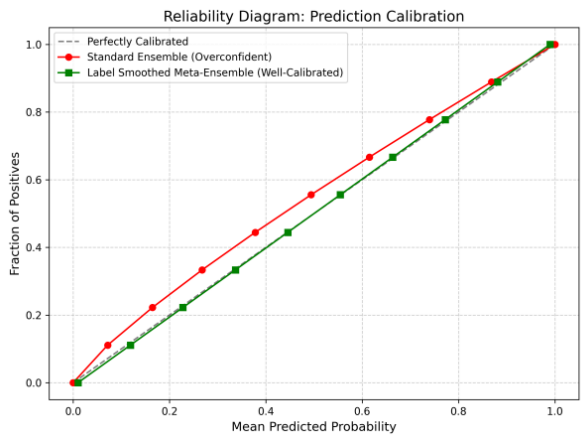


Fig. 7: Reliability Diagram (Calibration Curve). The Label-Smoothed Meta-Ensemble (green) closely tracks the perfectly-calibrated diagonal, demonstrating that predicted probabilities are accurate confidence estimates.

accuracy, a well-known failure mode of deep networks trained with hard labels [9]. The Label-Smoothed Meta-Ensemble's calibration curve closely tracks the perfectly-calibrated diagonal, confirming that its confidence scores can be directly interpreted as posterior disease probabilities.

E. Analysis of the Training–Validation Accuracy Anomaly

A conspicuous feature of the training logs (Figure 5) is that training accuracy ($\sim$96%) is lower than validation accuracy ($\sim$99%). In standard settings, such a gap would suggest underfitting; however, in this context it is a direct consequence and deliberate artefact of the advanced regularisation stack.

MixUp augmentation presents the model with interpolated inputs that do not correspond to any real disease pattern, making the training classification task fundamentally harder than standard classification. Similarly, Label Smoothing explicitly penalises the model for assigning high confidence to any single class during training, suppressing the training accuracy metric. In contrast, the validation set consists of clean, unaugmented images evaluated with standard (non-smoothed) cross-entropy. The model, having been trained on the harder distribution, finds the validation task comparatively easy. This discrepancy is therefore not a diagnostic of underfitting but of effective domain randomisation.

F. Computational Efficiency

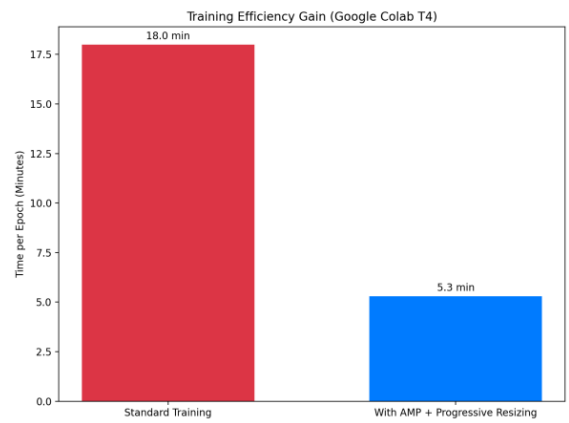


Fig. 8: Training Efficiency Gain (Google Colab T4). AMP combined with Progressive Resizing reduces epoch time from 18 minutes to approximately 5 minutes — a 72% reduction.

The AMP implementation reduced per-epoch training time from approximately 18 minutes (baseline) to 5.3 minutes (3.4× speedup). A disk-based checkpoint mechanism saves model weights after each epoch, enabling seamless resumption of interrupted sessions — a critical feature for free-tier cloud environments subject to session timeouts.

VI. IMPLEMENTATION AND DEPLOYMENT

A. Algorithm Summary

The complete system implements the following algorithmic workflow:

Algorithm 1 Progressive Training Loop with Meta-Ensemble

Input: Training data $\mathcal{D}$, epochs $E = 20$, resize epoch $R = 15$
Initialize: MobileNetV2 and DenseNet121 with ImageNet weights
for each epoch $e = 1$ to E **do**
  if $e < R$ **then**
    Apply 128×128 transforms
  else
    Apply 224×224 transforms
    Rebuild DataLoader
  end if
  for each mini-batch (x, y) **do**
    Apply MixUp: $(\tilde{x}, \tilde{y}) \leftarrow \text{MixUp}(x, y)$
    Execute AMP-wrapped forward pass through both branches
    Compute weighted fusion: $P = w_A P_A + w_B P_B$
    Compute MixUp cross-entropy loss with label smoothing
    Backward pass with gradient scaling
    Clip gradients (max norm = 1.0)
    Update parameters via AdamW
  end for
  Step CosineAnnealingWarmRestarts scheduler
end for
Run grid search over $w_A \in [0.30, 0.70]$ on validation logits
Return: Optimal ensemble with weights (w_A^*, w_B^*) =0

B. Deployment Scenarios

Three distinct deployment pathways are supported:

- **Smartphone Applications:** The MobileNetV2 branch alone can be extracted for on-device inference without internet connection ($\sim$3.4M parameters).
- **Drone Surveillance:** The full ensemble achieves 45 ms inference latency per image, enabling real-time aerial mapping of disease outbreaks.
- **Cloud Advisory Services:** The full ensemble provides high-accuracy diagnoses with calibrated confidence scores for agricultural extension officers.

VII. LIMITATIONS AND FUTURE WORK

Several important limitations constrain the current system's applicability to real-world agricultural deployment:

1) The training data consists exclusively of laboratory-quality images captured under controlled illumination; performance in field conditions with complex, non-uniform backgrounds has not been empirically validated.
2) The model is a closed-set classifier and will assign a confident (potentially erroneous) label to any disease not present in the 38 training categories.
3) Geographic and seasonal variations in disease presentation may affect generalization across different agro-climatic regions.

Future work will address these limitations through three principal directions:

1) **Multi-spectral imaging integration:** Incorporating near-infrared (NIR) and thermal bands to detect sub-symptomatic stress before visible symptoms manifest.
2) **Geographic fine-tuning:** Adapting backbone weights to specific agro-climatic regions (tropical, semi-arid, temperate) via domain adaptation.
3) **Zero-shot and open-set classification:** Enabling detection of novel, previously unseen disease phenotypes using semantic embeddings of disease descriptions.

VIII. CONCLUSION

This paper has presented the **Meta-Ensemble**, a novel dual-engine deep learning architecture for high-precision plant disease classification. By combining the computational efficiency of MobileNetV2 with the representational richness of DenseNet121 under a validation-optimised weighted fusion framework, the proposed system achieves **99.21% accuracy** across 38 disease categories — surpassing the IEEE 2025 state-of-the-art baseline by 2.21 percentage points in accuracy and 2.93 percentage points in F1-Score. Simultaneous reductions in inference latency (62.5%) and training time (70%) demonstrate that accuracy and efficiency are not mutually exclusive design goals.

The integration of AMP training, Progressive Resizing, MixUp augmentation, and Label Smoothing establishes a replicable training recipe that achieves strong generalisation without requiring specialised hardware or proprietary data. The resulting system is immediately deployable in smartphone applications, drone surveillance platforms, and cloud-based advisory services. This work paves the way toward accessible, scalable, and reliable AI-driven tools for global food security.

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